We will discuss different notions of equivalence between groups.

Quasi-Isometry

Definition: Recall that a map $f : X \to Y$ is called a *quasi-isometry* if there exists some constant $C > 0$ such that for all $x, y \in X$, we have that

$$\frac{1}{C} d_X(x,y) - C \leq d_Y(f(x), f(y)) \leq C d_X(x,y) + C$$

and there is some map $g : Y \to X$ and a constant K such that for all $x \in X$ and $y \in Y$, we have $d_X(g \circ f(x), x) \leq K$ and $d_Y(f \circ g(y), y) \leq K$. We write $X \sim_{\text{QI}} Y$.

Remark: We call a map $f : X \to Y$ such that there exist constants $C, B > 0$ with

$$\frac{1}{C} d_X(x,y) - B \leq d_Y(f(x), f(y)) \leq C d_X(x,y) + B$$

a (C, B)-quasi-isometric embedding.

Definition: A metric space (X, d) is said to be (C, B)-*quasi-geodesic* if there are constants $C, B > 0$ such that for any $x, y \in X$, there is a (C, B)-quasi-isometric embedding $\gamma : [t, t'] \to \mathbb{R}$ such that $\gamma(t) = x$ and $\gamma(t') = y$.

Example: If $X = (V, E)$ is a connected graph, then the graph metric on V turns V into a $(1,1)$-quasi-geodesic space.

When G is a finitely generated group, its word metric turns it into a $(1,1)$-quasi-geodesic space.

The Milnor–Schwarz Lemma

Proposition (Basic Milnor–Schwarz Lemma)**:** Let $G \curvearrowright (X, d)$ acts on a (C, B)-quasi-geodesic space. Suppose there is a subset $B \subseteq X$ such that the following hold:

(i) $\operatorname{diam}(B) < \infty$;

(ii) $\bigcup_{g \in G} g.B = X$;

(iii) the set $S = \{g \in G : g.B' \cap B' \neq \emptyset\}$ is finite, where

$$B' = \{x \in X : \text{ there exists } y \in B \text{ such that } d(x,y) \leq 2b\}.$$

Then,

(i) G is generated by S;

(ii) for all $x \in X$, the map $g \mapsto g.x$ is a quasi-isometry with respect to the word metric d_S on G.

References

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